Advanced Web Vulnerability Scanner using Python

This project demonstrates the development of an advanced Python-based web vulnerability scanner. The scanner is capable of detecting common cybersecurity weaknesses such as SQL Injection, Cross-Site Scripting (XSS), missing security headers, and exposed directories.

Project Objectives:
- Detect common web vulnerabilities
- Automate security scanning tasks
- Produce useful cybersecurity results
- Demonstrate Python applications in cybersecurity

Library	Purpose
requests	Sending HTTP requests
BeautifulSoup	Parsing HTML pages
urllib	URL processing
colorama	Colored terminal output
threading	Multi-threaded scanning
argparse	Handling command-line arguments

Tool Methodology:
1. Connect to the target website.
2. Check for missing security headers.
3. Test SQL Injection payloads.
4. Test XSS payloads.
5. Scan common hidden directories.
6. Extract links from the target page.

Python Code Example:

```
import requests
from bs4 import BeautifulSoup
from urllib.parse import urljoin
from colorama import Fore
```

Sample Results:
[+] Checking Security Headers...
[MISSING] Content-Security-Policy
[VULNERABLE] Possible SQL Injection Found
[FOUND] /admin directory

Results and Analysis:
The scanner successfully identified multiple security weaknesses, including: - Missing security headers - Potential SQL Injection vulnerabilities - Publicly accessible directories - Extracted links from the target website The project demonstrates how Python can automate practical cybersecurity tasks.

Conclusion:
This project demonstrated the development of an advanced web vulnerability scanner using Python. The scanner combines multiple cybersecurity techniques into a single tool and produces meaningful results.

References:
- https://owasp.org/
- https://docs.python.org/
- https://portswigger.net/web-security